2. Lego Mindstorms

Lego has already made a name for itself as one of the leading construction toys manufacturers around. The company took it a step further by announcing the 'Lego Mindstorms' project.

The main component of Lego MindStorms is the brick. This brick is the programmable brain of the entire kit. It consists of four input ports, four output ports, one mini-USB port to connect to a computer (by which users upload the code to the brick) and one extra USB port (to add Wi-Fi capabilities or anything else). It also comes with a micro-SD slot for file storage and a speaker. The other main components of the kit are mostly sensors that detect color, touch and angular velocity. Miscellaneous parts like wheels and connecting bricks are included as well. The brick can be used as a tool to nurture your child's programming skills. Lego offers its own proprietary software that lets users 'code' in a GUI interface with tiles, known as 'blocks', representing blocks of code. Each block has a certain color code. There are four main blocks: Action blocks, Flow blocks, Sensor blocks and Data blocks. There's an additional sub-block in blue, known as Advanced blocks, meant for connections via Bluetooth or Wi-Fi.

The Mindstorms kit can be used to create a list of robots recognised by Lego as its own creations, like the 'TRACK3r', 'SPIK3R' and so on. But it has also been extensively used in DIY creations. For example, 12-year-old California-based Shubham Banerjee developed a low-cost DIY braille printer when he found out that retail printers were extremely expensive. This single project has led to him starting his own company and developing a low-cost printer known as 'Braigo' for sale.

Lego Mindstorms: *http://dgit.in/LegoMindstorms*

3.Sphero

'Sphero' is just what it sounds like – a sphere. But this isn't just any sphere, you can control it with your smartphone. That's right, the ball-like robot can